

The Price of Eschewing Smoothness and Stopping Early: a Theoretical and Empirical Analysis of Conformal Methods

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Abstract

Conformal practice makes two choices so uniformly that they pass for definitions: residuals are pooled rather than smoothed, and the calibration step ends the chain. Of 686 arXiv papers titled with conformal prediction, 94.5% never depart from pooling at any granularity and 87% stop at the empirical map. We price the two choices separately, on 212 economic series against a strong online forecaster. Pooling costs 0.14 nats and buys the guarantee; a separation theorem shows local smoothing cannot lower that price without forfeiting exactness. Stopping costs a further 0.44 nats and buys nothing. Coverage detects neither.

1 Introduction

A conformal prediction system is assembled in a fixed order: fit a model, compute residuals on held-out data, pool their empirical law, read the answer off the pooled quantiles. Two conventions hide in that sentence. The residual law is *pooled*: one shape serves every state, however much the truth varies with the input. And the empirical map is *terminal*: nothing is fitted to its output. Neither convention is argued for in the papers that adopt it. This paper measures what each costs, and separates the price that buys the coverage certificate from the price that buys nothing.

The paper has exactly four components, and each subsequent section is one of them.

1. **The count** (Section 2, empirical). In a frame of 686 arXiv papers, 599 stop at the empirical map and 648 never depart from residual pooling at any granularity. One paper composes a model after the map. Every smooth-weighting paper that claims exactness was read in full; none overclaims, and the escapes they use are exactly the three the theory permits.
2. **The floor** (Section 3, a proposition). The excess logarithmic risk of any frozen conformal scheme equals a conditional mutual information fixed by its score and stratification. No calibration choice moves it. This is classical in spirit and stated because Section 5 uses it.
3. **The separation theorem** (Section 4, the theoretical centre). Exact distribution-free validity forces uniform pooling of the calibration residuals; test-centred smoothing opens a coverage gap of the same order as its bias, and the conflict is greatest exactly where smoothing would help most, at small samples. So the floor of Section 3 cannot be undercut from within.

4. **The price** (Section 5, empirical). Holding one strong forecaster and one empirical object fixed and moving only the object’s *position* in the chain: pooling costs 0.14 nats net of estimation effects, while stopping costs a further 0.44, three times more. The sign of the CRPS comparison flips across the same line. Adaptivity in time recovers 0.075 of it. Coverage registers none of it.

Section 6 states the recommendation, which is not a better calibrator: keep the map, keep the certificate where it is wanted, and keep modelling past it.

What is new and what is borrowed. The count, the two results, and the price are this paper’s; both results are proved here, the floor in Appendix A and the separation in Appendix B. The key coverage inequality behind the separation is due to Barber et al. (2023). The impossibility of exact distribution-free conditional coverage is Lei and Wasserman (2014) and Barber et al. (2021); the inflation forced by training-conditional coverage at a fixed point is classical (Li, 1989; Low, 1997; Cai and Low, 2004). The base forecaster is the subject of Cotton (2026).

2 The count

This section establishes that the two conventions are near-universal. It is a claim about published work, so it gets a protocol rather than a citation.

2.1 Frame and classification

The frame is every arXiv paper with “conformal prediction” in the title at harvest date, $n = 686$. Each paper is classified on two orthogonal axes by documented keyword screens over titles and abstracts, with the cells that matter verified by reading; classifier, frame, and per-paper verdicts are in the repository.

The position axis records what a paper does to the fence-post empirical map E_n : quote its output as the answer (STOP); substitute a conditional estimate for the pooled quantile (REPLACE); keep the map and adapt the level online (TUNE); feed its output to a downstream decision (CONSUME); or keep the map *and fit a model on its output* (COMPOSE), the only move that refutes the stopping claim.

The pooling axis records whether the weights on calibration residuals ever depend on the test point. POOLED means one empirical law; STRATIFIED means pooling within a pre-committed partition (Mondrian, binned, group-conditional), which coarsens the granularity but still pools; SMOOTH means test-centred weighting (localized, kernel, similarity weights on the calibration residuals). A smooth *score* is not smooth pooling: a kernel or normalized score is a richer score map, and the shape it reshapes is still one pooled object. The first version of the screen missed this distinction and was corrected; the correction is documented in the classifier.

2.2 Results

Three facts carry forward. First, 599/686 stop, and the single flagged compose candidate did not survive reading; the missing move of Section 5 is missing from the literature. Second, 648/686 (94.5%) never depart from residual pooling at any granularity. Third, the two axes are close to independent except in one corner: REPLACE and SMOOTH nearly coincide (14 of 16), as they

	POOLED	STRATIFIED	SMOOTH	total
STOP	523	55	21	599
REPLACE	2	0	14	16
TUNE	21	4	3	28
CONSUME	36	1	1	38
n/a	4	0	1	5
total	588	60	38	686

Table 1: Position of the empirical map against pooling of the calibration residuals, 686 papers, after full-text reclassification of two screen false positives. COMPOSE does not appear as a row because one candidate was flagged and none confirmed.

should, since substituting a conditional estimate is the smooth move. Twenty-one papers weight smoothly and still stop, paying both of the title’s prices at once.

2.3 The nine reads

Nine SMOOTH papers also claim exact or distribution-free validity in the abstract. Under the separation theorem of Section 4 each must be using one of three escapes, so all nine were read in full. None overclaims. Four restore *marginal* validity by recalibration or design (Guan, 2023; Hore and Barber, 2023); one states a coverage bound with an explicit gap in the style of Barber et al. (2023); one uses known covariate-shift tilts (Tibshirani et al., 2019); one proves finite-sample *local* bounds by assuming a smoothness class outright, surrendering distribution-freeness. Two were screen false positives whose smoothing lives in the score, and were reclassified POOLED. Hore and Barber (2023) state the boundary in the community’s own words: for local coverage, “there are no known distribution-free theoretical results to guarantee this.”

The field’s best localization papers therefore already respect the line the theorem draws. What the theorem adds is why the line is where it is, and Section 5 adds what crossing it, or declining to, is worth.

2.4 Refutation criteria

The stopping claim is falsified by a paper that retains the empirical rank transform, applies it to residuals, and fits a downstream model on the transformed stream, keeping the transform in the chain. Reweighting the empirical law does not count; conditioning upstream does not count; adapting the level does not count. The pooling claim is falsified by exhibiting exact distribution-free validity with test-centred weights outside the known covariate-shift escape. Outside the conformal banner the composing move is routine: the normal quantile transform and meta-Gaussian forecasting (Bogner et al., 2012), copula time series (Chen and Fan, 2006), and autoregressive transformation models (Rügamer et al., 2023) all apply an empirical marginal map and keep modelling in the transformed coordinates. The absence is a fact about a community’s habit, not about the difficulty of the idea.

3 The floor

This section states the identity that Section 5 measures against. It is borderline classical and is presented as a proposition with lineage, not as a discovery.

Fix a nonconformity score $A : \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R}$, strictly increasing in its second argument, and a stratification $\kappa : \mathcal{X} \rightarrow \mathcal{K}$, both frozen independently of the calibration labels. Every split, normalized, Mondrian, quantile, and predictive-system conformal scheme emits the same object: the pooled empirical law of the scores in the test point’s stratum, reshaped through $A(x, \cdot)^{-1}$; the named schemes differ only in (A, κ) . Write $W = A(X, Y)$.

Proposition 1 (Irreducible floor). *Suppose (A, κ) is frozen independently of the calibration labels, each $A(x, \cdot)$ is a strictly increasing absolutely continuous bijection, and the relevant densities and divergences exist and are finite. Then the excess logarithmic risk of the conformal predictor over the conditional oracle satisfies*

$$L(\widehat{F}_n) - H(Y | X) = I(W; X | \kappa) + \Delta_n, \quad \Delta_n \rightarrow 0,$$

where Δ_n is the pooled-density estimation term. Consequently no choice of calibration estimator, quantile convention, randomization, or level reduces the excess below $I(W; X | \kappa)$; only a change to (A, κ) can.

The proof is a change of variables in which the Jacobian of A cancels between truth and forecast, followed by the definition of conditional mutual information; it is reproduced in Appendix A. With a location score and one stratum the floor is the plain residual information gap $I(R; X)$. Level adaptation (Gibbs and Candès, 2021) and fold averaging act on the pooled scores alone, so they move Δ_n and nothing else.

Remark 2. The floor is zero exactly when (A, κ) renders the score conditionally independent of the input, that is, when the modelling was finished before the conformal step. Conformal calibration is deployed precisely when one is unwilling to assume that. The method’s stated reason for existing is its lower bound.

4 The separation theorem

This section is the theoretical centre. It shows the floor of Proposition 1 cannot be undercut from within: the one estimator that would lower it, local smoothing of the calibration residuals, is exactly the estimator that exact validity forbids.

Assume the conditional residual law μ_x is Hölder- s in total variation and is not input-free. The efficient estimator of μ_{x_0} at effective sample size m is kernel-weighted with bandwidth $h^*(m) \asymp m^{-1/(2s+d)}$, wide when m is small. The coverage of a weighted conformal set obeys the bound of Barber et al. (2023): localization costs, in coverage, the weighted total-variation distance between calibration and test residual laws, which under the Hölder condition is of order h^s .

Theorem 3 (Small-sample separation). *No weighted residual predictor attains both exact distribution-free coverage for every law in the class and the minimax estimation rate for μ_{x_0} , unless the sample is unbounded or the residual law is input-free. Exact validity forces the weights uniform within a pre-committed stratification; the estimation-optimal bandwidth incurs a coverage gap of order $h^*(m)^s$; and the product of the two deficits is bounded away from zero at every finite m .*

Corollary 4 (The scarce-data regime). *The separation is maximal as $m \rightarrow 0$: in the regime where local smoothing most improves the estimate, the exactly-valid predictor pays its full heterogeneity constant.*

Three consequences frame the measurement. First, STRATIFIED practice in Table 1 is rational: pooling within pre-committed strata is what the theorem permits, and it is how granularity can be bought without losing exactness. Second, the SMOOTH escapes found in Section 2.3 are the theorem’s own taxonomy realised: retreat to marginal validity, state the gap, assume the class, or move the smoothness into the score. Third, the known escapes are bounded. Fixed covariate-shift tilts (Tibshirani et al., 2019) evade the uniform-pooling conclusion because their weights do not depend on the residual laws; refining the stratification toward sufficiency is permitted but, because the labels may not steer the refinement, delivers a label-blind regressogram, minimax-suboptimal wherever adaptation would help (Vovk, 2022; Stone, 1982). Coverage at a fixed point under a training-conditional guarantee inflates by the local modulus for classical reasons (Li, 1989; Low, 1997), an orthogonal axis not used here.

What no result in this section touches is *position*. The theorem constrains how the pooled law may be estimated. It says nothing about whether the chain may continue past it, because position is not a coverage question. That is the gap the next section measures.

5 The price

This section holds one forecaster and one empirical object fixed and moves only the object’s position in the chain. The design answers one question: of the information a conformal-shaped step discards, how much is the floor of Proposition 1, which Theorem 3 protects, and how much is the stopping convention of Section 2, which nothing protects?

5.1 Design

The base forecaster is `laplace`, an online likelihood-weighted pool of roughly sixty candidate transform chains with multi-scale mixing, a generalized Pareto tail splice, and online PIT resolution; it is the strongest arm in the benchmark this study extends. The empirical object is a 41-node quantile grid of the base model’s own recent h -step residuals, kernel-smoothed at the Silverman bandwidth, the same construction used by the AutoARIMA-plus-conformal baseline in the public benchmark. Two arms place it:

stop. The conformal convention. Take the finished predictive, keep its location, replace its shape with the pooled empirical law. Nothing acts downstream. This is a frozen conformal scheme with a location score and one stratum, so Proposition 1 applies to it with floor $I(R; X)$.

compose-after. The same object installed as the terminal residual law *inside* the chain: the candidate pool still weights it, the tail splice still repairs its extremes, PIT resolution still runs, and the transform chain still inverts it to the observation coordinate. This arm is not a conformal scheme; its design is not frozen and it carries no certificate. It is the one move Proposition 1 permits, a change of representation, executed after the map instead of before it.

Windows of 250, 400, and 750 residuals are all reported. Recency-weighted variants (exponential weights, half-life 150) run in both positions, so no comparison can be attributed to staleness; by Theorem 3 the weighted variants have already surrendered exactness, which sharpens Section 5.4. The corpus is 212 FRED change series with at least 1,100 observations, context 1,024, twelve rolling origins per series, horizons 1, 6, and 12. Long contexts are necessary: at context 256 the three windows coincide and the arms collapse into one model with three names. Differences carry paired bootstrap intervals clustered by series, and win/draw/loss counts come from a per-series Diebold–Mariano test. A control corpus of iid series in three laws (Gaussian, t_3 , Laplace) runs first: there pooling is correct, and a constrained arm should lose nothing.

5.2 The decomposition

	pooled shape	stops	cost, net of estimation
unconstrained	no	no	—
compose-after	yes	no	0.14 nats
stop	yes	yes	0.58 nats

Table 2: Held-out log-likelihood deficit against the unconstrained base, $h = 1$, net of the iid control. The pooled shape costs 0.14; stopping costs a further 0.44, three times more.

The raw deficits at $h = 1$ are 0.846 nats for stop (window 400; 1.087 and 1.003 at windows 250 and 750) and 0.268 for compose-after. On the iid control the same arms lose 0.267 and 0.127: about a third of each raw gap is the cost of estimating 41 empirical quantiles rather than a parametric scale, and it is subtracted in Table 2. The control behaves: a state-correcting reference arm loses nothing there. At $h = 6$ the position term is 0.47 nats.

The theory mapping is asymmetric by construction. The stop arm is a genuine frozen conformal object, so its floor is $I(R; X)$ plus estimation. The compose-after arm is the permitted move, and its 0.14 is the residual gap of a far richer representation. The 0.44 between them is therefore a measured lower bound on how much of $I(R; X)$ a richer representation recovers: three quarters of the stop arm’s information loss was never floor at all. No theorem defends paying it, and by Table 1, 599 of 686 papers pay it.

5.3 The CRPS sign flip

arm	logpdf deficit	CRPS vs. base
stop, $w = 250$	−1.087	−1.4%
stop, $w = 400$	−0.846	+4.9%
stop, $w = 750$	−1.003	+4.9%
stop, recency-weighted	−0.771	−0.4%
compose-after, $w = 400$	−0.268	− 7.0%
compose-after, weighted	−0.275	− 8.1%

Table 3: $h = 1$, raw. Negative CRPS is an improvement over the unconstrained base. The same empirical object makes CRPS worse when placed last and better when placed early.

Placed last, the empirical law degrades CRPS by about five percent relative to the parametric leaf it replaced. Placed early, it improves CRPS by seven to eight percent, beating the base forecaster’s own CRPS-tuned leaf. Two proper scores disagree in sign across exactly the position line. This is the strongest form of the empirical claim, because CRPS is the score on which conformal practice rests its case, and the result cannot be dismissed as a log-score artifact.

5.4 Adaptivity, priced

Recency weighting moves the stop arm from -0.846 to -0.771 : adaptivity in time recovers 0.075 nats where position recovers 0.58. Adaptive weighting tracks the level of volatility through time and still quotes one shape per step regardless of state; the state-dependent version that would close the gap is what Theorem 3 prices. Level adaptation is dispatched by Proposition 1 directly, since it acts on the pooled scores alone.

5.5 The certificate is blind to the loss

arm	marginal	falling vol	flat	rising vol	spread
unconstrained	0.782	0.814	0.741	0.823	0.082
stop, $w = 400$	0.819	0.863	0.782	0.789	0.080
compose-after, $w = 400$	0.810	0.867	0.785	0.801	0.082

Table 4: Empirical 80%-interval coverage at $h = 1$, marginally and within volatility regimes. The arm losing 0.85 nats has the best-looking coverage.

We expected the stop arm to show good marginal and poor conditional coverage. It does not: its calibration is the best in the table, marginally and within regime. No claim of miscalibration can be made against it, and none is. The loss is information, which coverage does not measure. ? states the reason in advance: the certificate concerns the marginal rank of the test score, the floor concerns the conditional information the frozen score discards, and the two live on different axes. Table 4 is that statement, verified: we measured the axis the certificate lives on, and it registered nothing.

6 Discussion

The two conventions of the title now have separate prices and separate statuses. Eschewing smoothness costs 0.14 nats here. It is protected: Theorem 3 shows the exact guarantee cannot be kept without it, and the nine reads of Section 2.3 show the field’s own localization literature already respects that boundary. Stopping early costs 0.44 nats here, three times more. It is protected by nothing, defended nowhere, and practiced in 87% of the frame.

The recommendation is therefore not a better calibrator, a smarter weighting, or a tuned level; Proposition 1 closes that entire menu. It is to treat the empirical map as what it provably is, a coordinate change with a certificate attached at its position, and to keep modelling on its output. The compose-after arm is one constructive existence proof, and it is conformal prediction with the modelling finished, in the sense of Remark 2 run in reverse: the certificate is what one holds when the modelling is not finished, and the PIT stream of the composed chain measures, rather than assumes, whether it ever was.

Limitations. The count is a keyword screen with full-text verification of the cells that carry the argument: the smooth-with-exact-claim cell is fully read, the compose shortlist is fully read, and the 599-paper STOP default rests on the screen plus a documented refutation criterion. Every price is measured on one base forecaster, one economic corpus, and one iid control; the position effect should be replicated on a second base before the constants are quoted as more than one instance. The compose-after arm carries no coverage guarantee, which is the point being priced, and it must not be read as a proposed better conformal method.

A Proof of Proposition 1

Fix the fold, so (A, κ) is deterministic. Given $X = x$, the true density of Y is $p(y|x) = p_W(A(x, y) | x) A'(x, y)$ by change of variables, and the conformal density is $\hat{f}_x(y) = \hat{g}_{\kappa(x)}(A(x, y)) A'(x, y)$. The Jacobian cancels in the log ratio, so $\text{KL}(P(\cdot | x) \| \hat{F}_x) = \text{KL}(P_W(\cdot | x) \| \hat{G}_{\kappa(x)})$. Splitting at the population pooled law $G_{\kappa(x)}$ and averaging over X gives $\mathbb{E}_X \text{KL}(P_W(\cdot | X) \| P_W(\cdot | \kappa(X))) + \Delta_n$, and the first term is $I(W; X | \kappa)$ by definition. The floor depends only on (A, κ) ; every calibration operation is a function of the pooled scores and so moves only Δ_n . $\square$

B Proof of Theorem 3

Write the weighted estimator $\hat{\mu}_w(x_0) = \sum_i w_i(x_0) \delta_{R_i}$ with kernel weights of bandwidth h . Two displays give the two horns.

Localization opens a coverage gap. By the coverage bound for non-exchangeable weighted conformal prediction (Barber et al., 2023), the level- $(1 - \alpha)$ set of the weighted law satisfies

$$\Pr\{R_0 \in C_\alpha(x_0)\} \geq 1 - \alpha - \sum_i \tilde{w}_i(x_0) d_{\text{TV}}(\mu_{X_i}, \mu_{x_0}),$$

attained in the worst case over laws with the stated total variations. Under the Hölder- s condition the weighted sum is $\leq C \sum_i \tilde{w}_i \rho(x_0, X_i)^s = \Theta(h^s)$, since kernel weights concentrate on $\{\rho \lesssim h\}$.

Exact validity forces uniform pooling. Distribution-free validity for every law in the class forces the localization term to zero. It is a weighted average of nonnegative total variations, some strictly positive under heterogeneity, so the weights must place no differential mass across covariate-distinct calibration points: uniform pooling within a partition committed before x_0 is seen, the $h \rightarrow \infty$ endpoint.

The separation. Uniform pooling has total-variation bias equal to the heterogeneity constant of the pooled region, positive and independent of m , violating the minimax rate $m^{-s/(2s+d)}$, which tends to zero. Conversely any estimator achieving that rate uses $h \asymp h^*(m) = m^{-1/(2s+d)}$ and by the first display has coverage gap $\Theta(h^*(m)^s) = \Theta(m^{-s/(2s+d)})$, positive at every finite m . Bias and coverage gap are both $\Theta(h^s)$ along the bandwidth family, so the product of the two deficits is bounded away from zero for finite m and vanishes only as $m \rightarrow \infty$. The corollary is the monotonicity of $h^*(m)$ in m . $\square$

Reproduction. Study declarations, the classifier with its documented corrections, per-paper verdicts, and the tagged result store are in the public repositories; every table above regenerates from them.

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